Scalability & Future Plan

Date	03 July 2026
Team ID	
Project Name	**Credit Card Approval Prediction**
Maximum Marks	1 Mark

Scalability & Future Plan: This document describes how the **Credit Card Approval Prediction** system can be scaled to support more users, larger datasets, and advanced features in the future. It also presents the roadmap for enhancing the project beyond its current implementation

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited historical applicant dataset	May reduce prediction accuracy for diverse applicants	High
2	Local model deployment	Difficult to support many concurrent users	Medium
3	Basic security implementation	Requires stronger authentication and secure data handling	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy the application on IBM Cloud with load balancing to support thousands of users.
2	Data Storage	Dataset stored locally in CSV format	Store applicant records in a cloud database such as MySQL or PostgreSQL.
3	Performance	Single-server Flask deployment	Optimize the model and deploy using Gunicorn/Nginx or cloud services for faster predictions.
4	Security	Basic input validation	Implement user authentication, HTTPS encryption, secure APIs, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase1	Deploy the Credit Card Approval Prediction system on IBM Watson Machine Learning and IBM Cloud.	Next 1 Month	Public accessibility and scalable cloud deployment.
Phase2	Add user login, applicant history, and prediction tracking.	Next 2 Month	Improved security and better user management.
Phase3	Integrate real-time credit score and banking APIs for enhanced decision-making.	Next 3 Month	More accurate and reliable credit card approval predictions.
Phase4	Develop a mobile application, multilingual support, and AI-based recommendation features.	Next 4 Month	Better accessibility and wider user adoption